BASANT BHATTARAI

DATA & AI ARCHITECT / SENIOR DATA ENGINEER

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PROFILE

Data and AI architect and senior data engineer with 9+ years on production systems in industry. Sets the target architecture, chooses platforms on measured evidence, designs the data model and the governance and SOC 2 controls around it, then leads the build. Builds on open-source engines — Kafka, Spark, Iceberg, Trino, dbt and Airflow on Kubernetes — and the databases underneath: modeling, replication, backup and recovery, upgrades. AI built into the platform: agent workflows on LangGraph, Google ADK and MCP whose output lands as typed, reviewable rows, with isolation, evaluation and review designed in. Track record: 40% lower infrastructure cost, 50% faster processing, 60% faster analytics delivery.

EXPERIENCE

Senior Data Engineer | UXCam

Feb 2024 - present

- Own the architecture, reliability and governance of a platform handling 15 TB+ and 100M+ events daily: freshness and latency objectives per dataset, 99.9% uptime, incident-ready runbooks, reviewable data contracts, a named owner for every published table, and the SOC 2 criteria met for the platform and its infrastructure through every audit.
- Design and operate the production agent workflows — LangGraph and Google ADK orchestration, MCP tool servers, retrieval and evaluation — built into the existing platform, so model output lands as typed, provenanced rows rather than a transcript: analytics delivery 60% faster with 75% less analyst effort, review and fallback paths kept.
- Operate the SQL and NoSQL databases underneath: replication, backup and recovery, access, tuning, upgrades.
- Lead architecture and schema reviews before new stores, pipelines and agent workflows reach production.
- Mentor 5 engineers through pairing, code review and written design feedback; team delivery time down 30%.

Data Engineer | UXCam

Feb 2020 - Feb 2024

- Rebuilt the core ETL layer in Spark and PySpark with Airflow orchestration, cutting processing time for 5 TB+/day by 50% and unblocking real-time product dashboards.
- Made schema evolution, replay, late data and backfills designed interfaces: idempotent loads, explicit watermarks.
- Tuned Trino, ClickHouse, Citus and TimescaleDB serving 100M+ queries/day for a 50% p95 latency reduction through partitioning, clustering and model redesign.
- Led the Kubernetes migration of the data infrastructure: 40% lower infrastructure cost, 99.9% uptime, and autoscaling that held under peak load.

Independent Consultant | Data, databases and AI

May 2021 - present

- 30 engagements alongside UXCam. Database architecture and migration: Postgres scaled out to Citus, RDS moved to a self-managed Citus cluster, a 500 GB database moved and faster afterwards, a ClickHouse cluster designed to its workload.
- PostgreSQL and TimescaleDB clusters with replication, pgBackRest and PgPool-II; a database recovered from raw files; AWS database cost analysis; an analytics lakehouse on S3, Trino, Airflow and Glue, end to end.
- Frameworks for LLM applications and agentic workflows, from the architecture to tested Python; a short-text classifier redesigned on Databricks; pgvector performance on RDS.

Project Leader | SVCET, India

2019 - 2020

- Led 4 developers building an intelligent dialogue system on an NLP and deep-learning stack, reaching the client's acceptance threshold for intent recognition and delivering 2 weeks ahead of schedule.

Software Developer | SV Technology, India

2017 - 2019

- Built Python and FastAPI backend services carrying 10K+ users at 85% test coverage.
- Designed and maintained PostgreSQL schemas, stored procedures and reporting paths, and set up CI with Jenkins alongside early pipeline work, the start of the move into data engineering.

SELECTED OPEN SOURCE

- **data-platform-architecture** and **ai-platform-architecture** (reference architectures): a SaaS company leaving Postgres, measured: 18 decision records, a Kimball model, SOC 2 readiness, a cost model and a Databricks proof of concept; then its usage assistant on governed data.
- **spark-sql-kinesis** (maintainer): Kinesis source and sink for Spark 4.2 on AWS SDK v2, on Maven Central; an idle shard costs one GetRecords call, not a multi-second floor per batch.
- **distributed-db-lab**: Citus sharding, online rebalance and tenant isolation, measured against TimescaleDB and replicated ClickHouse. Also evalproof, data-stack-lab, a governed agent, streaming, real-time ML, retrieval, data contracts and GitOps delivery. github.com/technobasant

CAPABILITY MAP

Data and AI architecture

Platforms chosen on evidence, with every decision written down and AI built in.

Target architectures and platform selection, conceptual to physical modelling, Kimball and Type 2 history, lakehouse, ClickHouse and Citus layouts, contracts in CI, decision records, SOC 2 and erasure by design, tenant isolation outside the model, semantic layers, release gates, cost models.

Data platforms

Consumer-facing reliability, governance, recovery and operational readiness.

Python, SQL and Scala; Kafka and Kinesis ingestion (maintainer of an open-source Spark 4 Kinesis connector); Spark/PySpark processing, including the Spark 4 migration; Airflow orchestration; Iceberg, Hive Metastore, dbt and lakehouse modeling; Databricks with Unity Catalog; Kubernetes, Docker and Terraform delivery; contracts, lineage, cost and recovery; Prometheus and Grafana.

AI systems

Model output that lands as typed, reviewable rows rather than a transcript.

LangGraph, LangChain, Google ADK, MCP servers, Pydantic, Milvus and MLflow; bounded tools, retrieval, evaluation before rollout, provenance, human review and honest fallback paths.

Database lifecycle

Modeling, replication, failover, backup and PITR, upgrades and query performance.

PostgreSQL, ClickHouse, Trino, Citus, TimescaleDB, Redis and MongoDB; partitioning, replication, restore and high availability.

Cloud

AWS: S3, EKS (IRSA), Lambda, Glue, Athena, Redshift, Kinesis, EMR | **GCP:** BigQuery, Dataflow, Dataproc, Datastream, Pub/Sub, Cloud Composer, GKE, Vertex AI | **Azure:** Synapse, Data Factory, Databricks, ADLS, Event Hubs, AKS, Azure ML

EDUCATION AND LANGUAGES

Bachelor of Technology in Computer Science and Engineering

JNTUA College of Engineering, Anantapur, India | 2015 - 2019

English: C1 | Nepali: native

Selected work, writing and case studies: basantbhattarai.com.np/work/